# Regim dinamic

## Suprapozitie

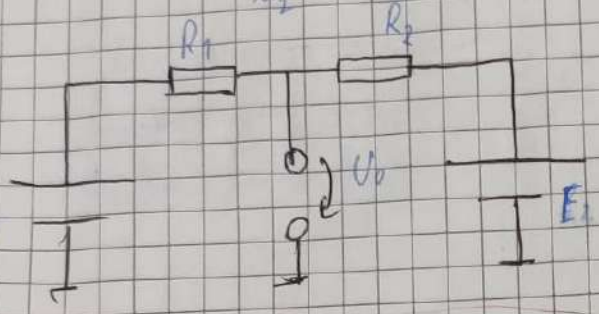
- dacă avem circuit liniar (cu componente liniare):

(funcție liniară $f(\alpha u_1 + \beta u_2) = \alpha f(u_1) + \beta f(u_2)$)

$f(u_1, u_2) = f(u_1, 0) + f(0, u_2)$

anulăm $u_2$ | anulăm $u_1$

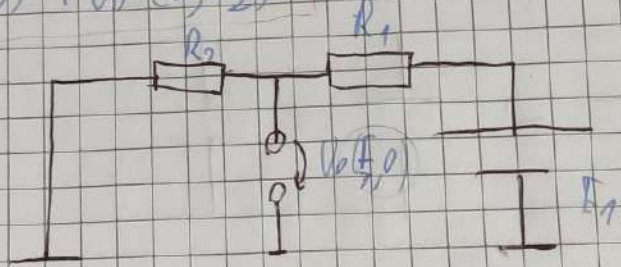
E.g.:




$U_0 = ?$

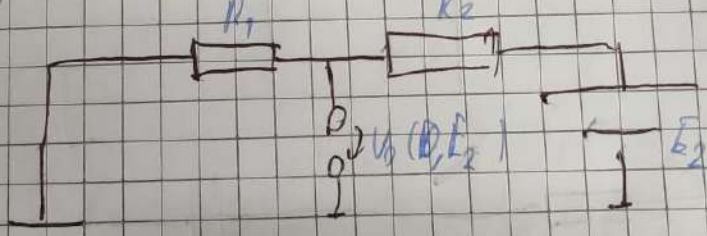
$U_0(E_1, E_2) = U_0(E_1, 0) + U_0(0, E_2)$

Pentru $U_0(E_1, 0)$:




$U_0(E_1, 0) = \frac{R_2}{R_1+R_2} \cdot E_1$ (divizor de tensiune)

Pentru $U_0(0, E_2)$:



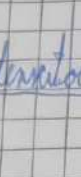

$U_0(0, E_2) = \frac{R_1}{R_1+R_2} \cdot E_2$

$\Rightarrow U_0(E_1, E_2) = \frac{R_2 E_1}{R_1+R_2} + \frac{R_1 E_2}{R_1+R_2} = \frac{R_2 E_1 + R_1 E_2}{R_1+R_2}$ (sursă reală legate în paralel)

2) Figura de unde de intrare:



- Când le avem pe amândouă, nu putem calcula prima efectiv simultan, astfel nu putem calcula pe rând

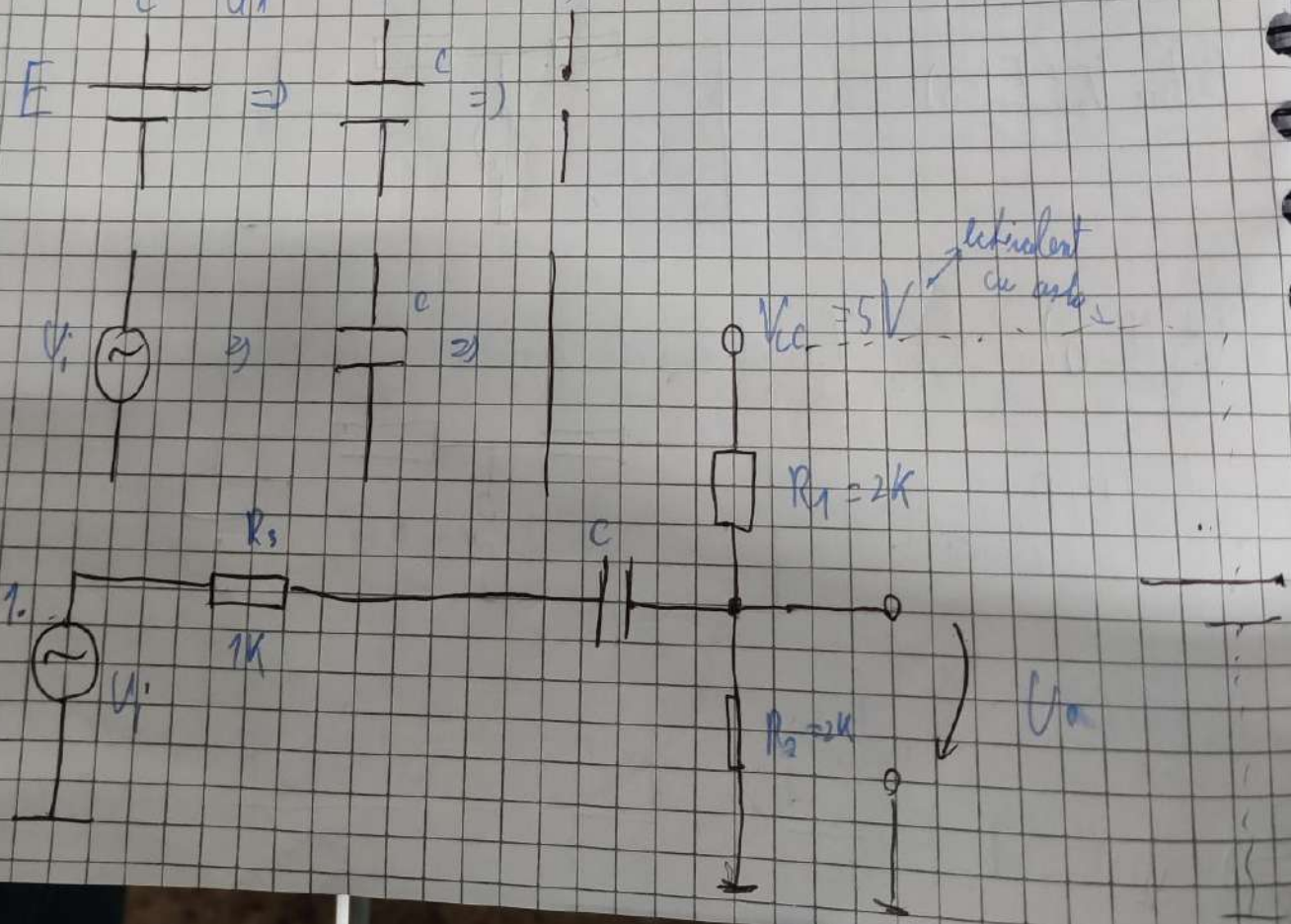
II) Condensatoare:  $U_C$ : $I_C = C\frac{dU_C}{dt}$

$I_C > 0 \Rightarrow U_C$ crește

$I_C < 0 \Rightarrow U_C$ scade

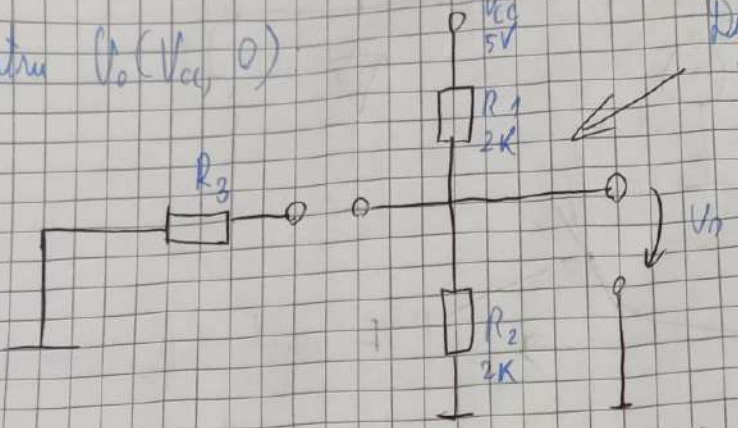
$I_C = 0 \Rightarrow U_C$ constant

$\frac{I_C}{C} = \frac{dU_C}{dt}$ (pentru C mare, U-variație mică)



$U_0(V_{cc}, U_i) = U_0(V_{cc}, 0) + U_0(0, U_i)$

Pentru $U_0(V_{cc}, 0)$

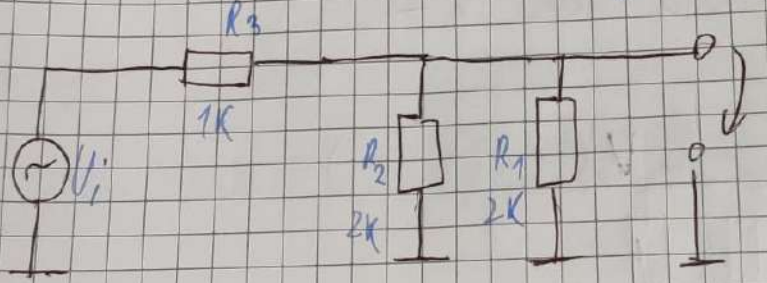



Divizor de tensiune

$U_0(V_{cc}, 0) = \frac{R_1}{R_1+R_2} \cdot V_{cc} = \frac{2K}{4K} \cdot 5V = \underline{2,5V}$

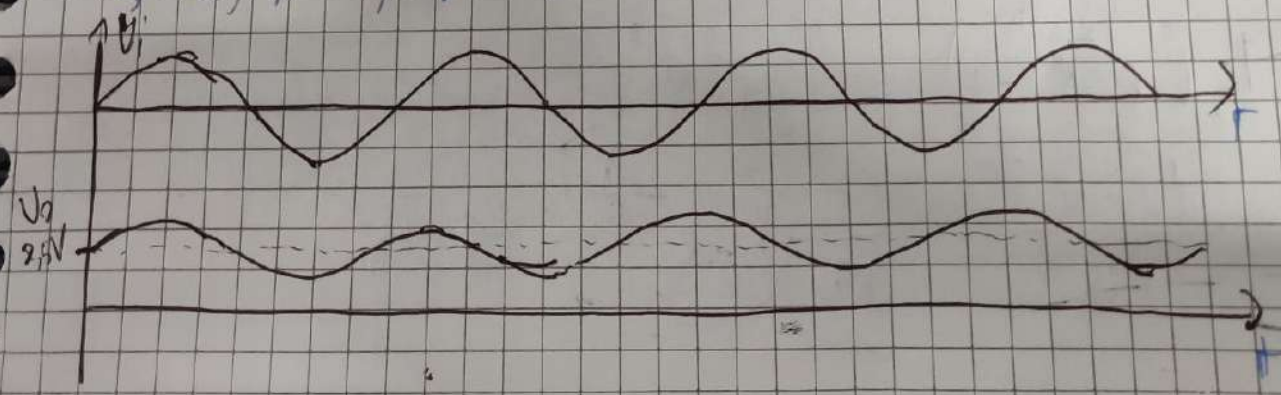
Pentru $U_0(0, U_i)$

$(R_2 \| R_1)$ și $R_3$ - Divizor de tensiune



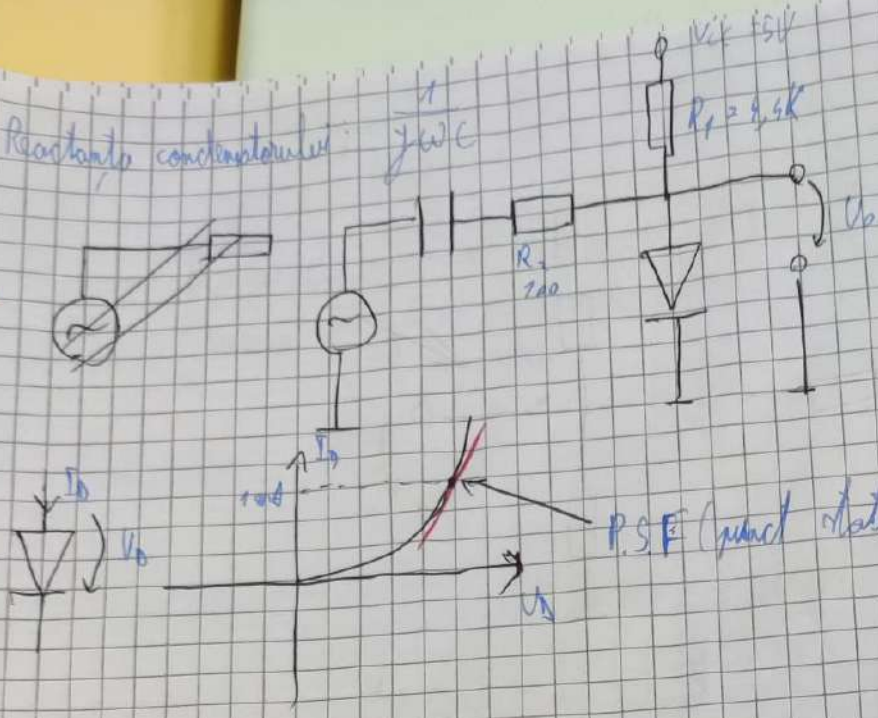

$U_0(0, U_i) = \frac{R_1 \| R_2}{R_3 + R_1 \| R_2} U_i = \frac{1K}{1K+1K} \cdot U_i = \frac{U_i}{2}$

$\Rightarrow U_0(V_{cc}, U_i) = 2,5V + \frac{U_i}{2}$



Reactanța condensatorului: $\frac{1}{j\omega C}$

2.




Algoritm pentru circuit neliniar

Regim Dinamic

- Aproximația Semnal Mic -

Algoritm

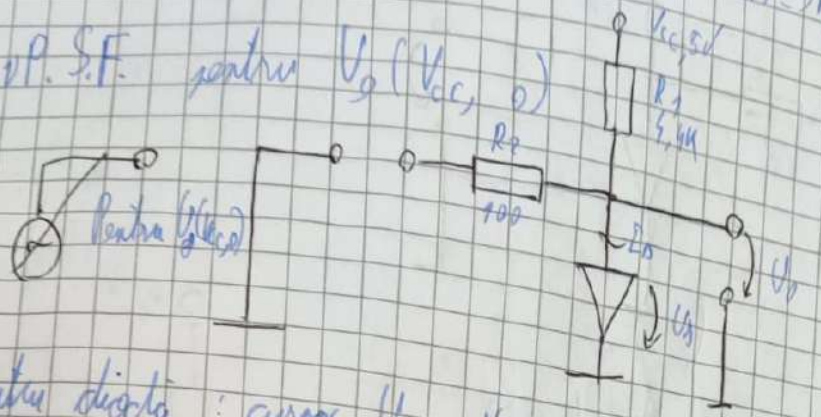
1. Calculăm P.S.F pentru $V_0$: $V_0(V_{cc}, 0)$
2. Liniarizăm în jurul P.S.F
3. Calculăm $V_0(0, V_i)$ în circuitul liniarizat
4. Aplicăm suprapoziția: $V(V_{cc}, V_i) = V_0(V_{cc}, 0) + V_0(0, V_i)$

$$I_D = I_S\left(e^{\frac{q_0 V_D}{KT}} - 1\right) \simeq I_S e^{\frac{q_0 V_D}{KT}}$$ curentul prin diodă

$$\frac{dI_D}{dV_D} = \frac{d\, I_S e^{\frac{q_0 V_D}{KT}}}{dV_D} = \frac{q_0}{KT}\left(I_S e^{\frac{q_0 V_D}{KT}}\right) = \frac{q_0}{KT} \cdot I_D$$

$\Rightarrow R_D = \frac{dV_D}{dI_D} = \frac{kT}{q_0 I_D} \simeq \frac{1}{40 I_D}$ , $I_D(mA) \rightarrow R_D(K\Omega)$

Rezolvă S.F. pentru $U_0(V_{CC}, 0)$




Pentru diodă: aprox $U_D = 0,6V$

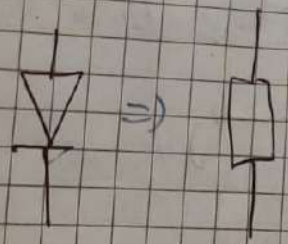
Presupunem: D - conductie

$U_D = 0,6 = U_0$

$V_{CC} = U_D + R_1 I_D \Rightarrow I_D = \frac{V_{CC} - U_D}{R_1} = 1mA$

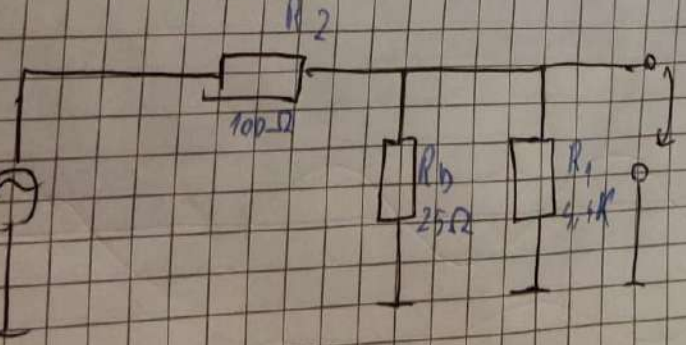
$\Rightarrow U_0(V_{CC}, 0) = 0,6V$

Pas 2) Limitare



$\Rightarrow R_D = \frac{1}{40 I_D} = 25\Omega$

Pas 3) Pentru ~~$U_0(V_{CC}$~~ $U_0(0, V_i)$



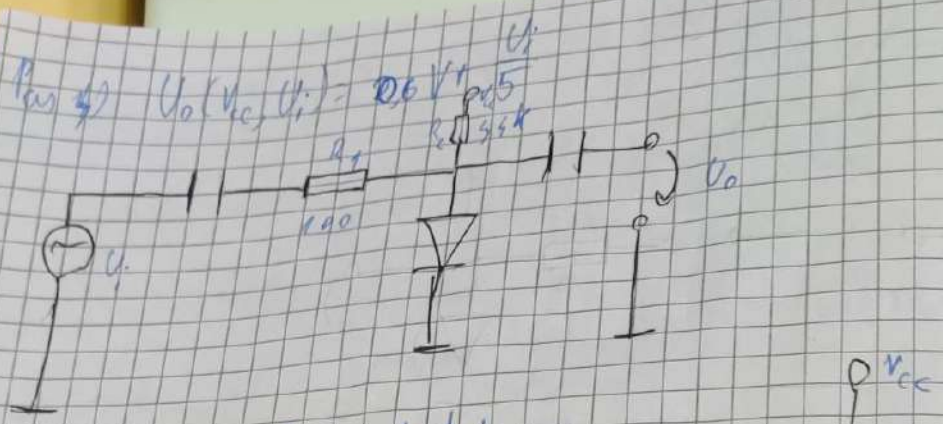

$R_1 \| R_D = \frac{25 \cdot 4400}{25 + 4400}$

$R_1 \| R_D \simeq 25$

$U_0(0, V_i) = \frac{R_1 \| R_D}{R_1 \| R_D + R_2} \cdot V_i \simeq \frac{25\Omega}{25\Omega + 100\Omega} \cdot V_i = \frac{V_i}{5}$

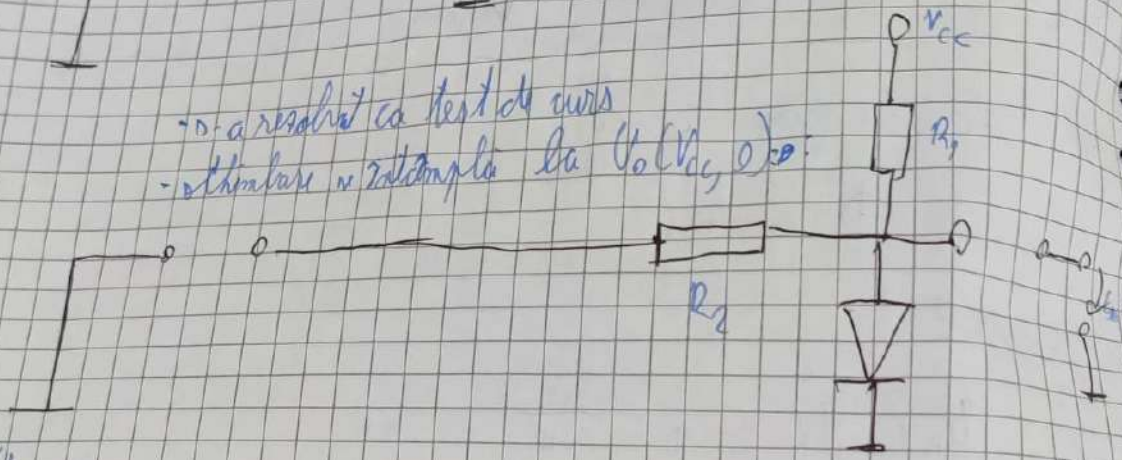
Rezolvare

2.



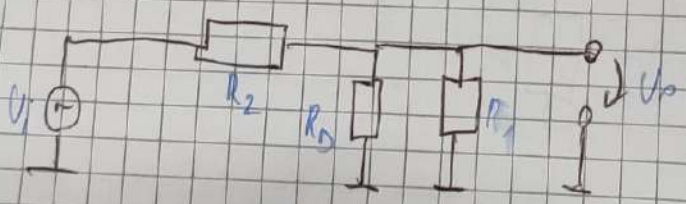

a) $U_o(V_{cc}, U_i) = 0,6 V + \frac{U_i}{5}$

- a rezolvat ca test de curs
- obținut o rezistență la $U_o(V_{cc}, 0)$:




$\Rightarrow U_o = 0$

- la $U_o(0, U_i) = \frac{U_i}{5}$ - NU se schimbă Nimic




$\Rightarrow U_o(V_{cc}, U_i) = \frac{U_i}{5}$